

MSCF Probability – Final Exam – August 24, 2020

Name _____

Email _____

Instructions:

- You should complete all questions on this exam by writing out the solutions on the paper which you will scan in the end.
- You will have three hours to complete this exam. No matter when you start, the exam will be due at a time to be specified, three hours after you start.
- Notes (except described below), books and calculators/laptops/phones are not allowed, except for the purposes of accessing, scanning, and submitting the exam.
- A collection of math facts, and a table of common distributions, with pdfs/pmfs, and other information, is included at the end of this exam.
- Final answers can involve any of the following notation:

$$\binom{n}{k}, n^k, \exp(k), n!, \Gamma(x), \Phi(x),$$

where $\Phi(x)$ is the CDF of the standard normal distribution. Except for cases in which it is explicitly stated, final answers should **not** include integrals.

- You may use 2 sides of a 8.5" by 11" page (1 double-sided or 2 single-sided) of notes.
- Questions are not all worth the same number of points. The numbers in **bold** after each question indicate the number of points. Also, you may find some easier than others. You should prioritize your work on particular questions accordingly.

Note: The total number of points is 90.

- You **must show all of your work, except for the multiple choice questions**. A correct answer without explanation will receive only minimal or no partial credit. Note that the “easier” a question may seem, the more burden you have to show that you understand how you are reaching the answer; again, **be clear and complete in your work**.

**PLEASE CHECK THAT YOU HAVE PAGES 1 – 8,
THE DISTRIBUTION TABLE, AND THE MATH FACTS**

Grading Guidelines:

The following are a few reasons why you may lose points:

1. Failure to mention when an independence assumption is utilized.
2. Failure to mention when the assumption that random variables are uncorrelated is utilized.
3. Failure to specify clearly the support of a distribution.

You do **not** need to cite the names of rules and definitions when you use them, such as

1. The Rule for the Lazy Statistician
2. The Iterated Expectation Rule
3. The Rule for Conditional Probability
4. The Addition Rule
5. The Chain Rule
6. ...

But, in general, any work that does not make it clear how the answer was found is subject to losing points. In other words, even if you do not justify every step, there should be sufficient number of steps to make it clear you know how to do the derivation. Lack of clarity can also result from messy writing: write neatly.

1. Suppose that A and B are **disjoint** events (i.e., $A \cap B = \emptyset$), and that $P(A) > 0$ and $P(B) > 0$. Which of the following statements **must** be true? (4)

(YOU CAN CHOOSE MORE THAN ONE.)

- (a) A and B are dependent (not independent). MUST BE TRUE
- (b) $P(A \text{ or } B) = P(A) + P(B)$ MUST BE TRUE
- (c) $P(A \text{ and } B) = P(A)P(B)$
- (d) $P(A \text{ and } B) < P(A)P(B)$ MUST BE TRUE

2. Suppose that X has the exponential(λ) distribution. Suppose that a and b are any positive constants. Which of the following are implied by the *memoryless property* of the exponential distribution? (4)

(CHOOSE ONE)

- (a) $P(X > a + b | X > a) = P(X > a)$
- (b) $P(X > a + b | X > b) = P(X > a) / P(X > b)$
- (c) $P(X > a + b | X > b) = P(X > a)$ CORRECT
- (d) $P(X > a + b | X > b) = P(X > b)P(X > a)$

3. Suppose that X has pdf given by

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right), \quad -\infty < x < \infty, \quad b > 0.$$

The moment generating function (MGF) for X is

$$M_X(t) = \frac{1}{1 - b^2 t^2} \quad \text{for } -1/b < t < 1/b$$

Note: X is said to have the *double exponential distribution* or *Laplace distribution*.

Suppose that Y_1 has the exponential(λ_1) distribution, and Y_2 has the exponential(λ_2) distribution, where λ_1 and λ_2 are positive constants. Assume Y_1 and Y_2 are independent. Define $Z = \lambda_1 Y_1 - \lambda_2 Y_2$.

Show that Z has the double exponential distribution with $b = 1$. (4)

Get the MGF of the exponential distribution from the table, and then apply properties of the MGF to get the result.

4. Suppose that events arrive as a Poisson Process with rate λ . (λ is in units of minute^{-1} .) Let X denote the number of events during two minute period.

What is the variance of X ? **(3)**

2λ

5. Suppose that (X, Y) has joint pdf given by

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{6}(x + 4y), & \text{if } 0 < x < 2, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Show that the marginal for Y is

$$f_Y(y) = \begin{cases} \frac{1}{3}(1 + 4y), & \text{if } 0 < y < 1 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

- (b) What is the probability that $X \leq 1$ given that $Y = 0.25$? **(5)**

$3/8$

6. Assume that X is a random variable, and $\mathcal{F}$ is a σ -field. Consider the following statement:

$$\sigma(E(X|\mathcal{F})) \subseteq \mathcal{F}$$

Which of the following do you agree with? **(3)**

(CHOOSE ONE)

- (a) This statement is definitely true. **CORRECT**
- (b) This statement is incorrect, because $E(X|\mathcal{F})$ is not a random variable.
- (c) This statement is incorrect, because $\sigma(E(X|\mathcal{F}))$ consists of subsets of $\mathcal{F}$, it cannot itself be a subset of $\mathcal{F}$.
- (d) There are circumstances under which this statement is true, and situations where it is false.

7. There are two boxes. Box 1 contains four red balls and two white balls. Box 2 contains three red and three white balls. One ball is drawn from Box 1, hidden from view, and then placed in Box 2 (so that Box 2 now has seven balls). Then a ball is drawn from Box 2. (You can assume that whenever a ball is drawn from a box, each ball in the box is equally likely to be drawn.)

(a) What is the probability that the ball drawn from Box 1 will be red? **(3)**

$2/3$

(b) What is the probability that the ball drawn from Box 2 will be red? **(4)**

$11/21$

(c) Given that the ball drawn from Box 2 is red, what is the probability that the ball transferred from Box 1 to Box 2 was red? **(4)**

$8/11$

(d) What is the probability that both balls drawn (the ball from Box 1 and the ball from Box 2) are red? **(4)**

$8/21$

8. Suppose that events of Type I arrive as a Poisson process with rate λ_1 , and that events of Type II arrive as a Poisson process with rate λ_2 . These two processes are independent: The occurrence of events of Type I has no effect on the occurrence of events of Type II, and vice versa. Also, λ_1 and λ_2 are in the same units.

Choose some arbitrary time point to serve as the “starting point.” Recall that due to the memoryless property of the exponential distribution, this choice is irrelevant.

What is the expected value of the number of events of Type I prior to the first event of Type II? **(3)**

Use iterated expectation. Answer is λ_1 / λ_2 .

9. A fair coin is tossed until the first “heads” appears. Let X denote the number of tails prior to that first heads. What is the distribution for X ? (4)

(CHOOSE ONE)

- (a) X is binomial($n = 2, p = 0.5$).
- (b) X is Geometric with $p = 0.5$. **CORRECT**
- (c) X is Poisson($\lambda = 0.5$).
- (d) X is Bernoulli with $p = 0.5$.

10. Which of the following can be proven regarding general random variables X and Y ? (4)

(CHOOSE ONE)

- (a) X and Y independent implies X and Y are uncorrelated. **CORRECT**
- (b) X and Y uncorrelated implies X and Y are independent.
- (c) X and Y independent if and only if X and Y are uncorrelated.

11. Suppose that A, B are events and that $P(A) = 0.4$ and $P(B) = 0.3$. What can be said about $P(A \cup B)$? (4)

(CHOOSE ONE)

- (a) $P(A \cup B)$ equals 0.70.
- (b) $P(A \cup B)$ equals 0.12.
- (c) It is possible that $P(A \cup B)$ is as small as 0.40 or as large as 0.70, but it cannot be outside that range. **CORRECT**
- (d) It is possible that $P(A \cup B)$ is as small as 0.30 or as large as 0.70, but it cannot be outside that range.
- (e) It is possible that $P(A \cup B)$ is as small as 0.30 or as large as 0.40, but it cannot be outside that range.

12. Assume X is a random variable which has the binomial(n, p) distribution. Note that X/n equals the proportion of the binomial trials which are successes. What are the mean and variance of X/n ? **(4)**

$E(X/n) = p$ and $V(X/n) = p(1 - p)/n$.

13. Let X be multivariate normal: $X \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & -0.3 & 0 \\ -0.3 & 1 & 0.2 \\ 0 & 0.2 & 2 \end{bmatrix}.$$

- (a) What is the joint distribution of (X_1, X_2) ? **(3)**

(X_1, X_2) are bivariate normal with mean $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ and variance $\begin{bmatrix} 1 & -0.3 \\ -0.3 & 1 \end{bmatrix}$.

- (b) Are X_1 and X_2 independent? Why or why not? **(3)**

They are dependent because they are correlated.

14. A box contains 5 **unfair** coins and each has a different probability of showing heads. Let p_i denote the probability of heads for coin $i = 1, 2, \dots, 5$. Suppose that $p_i = (i - 1)/4$. Suppose a coin is selected at random (each is equally likely to be chosen), and then that coin is tossed.

Let H denote the event “the coin toss results in heads” and let C_i be the event that coin i is selected.

- (a) The events $C_1, C_2, \dots, C_5$ are...

(CHOOSE ONE) (3)

- i. ...independent.
- ii. ...a partition of Ω . **CORRECT**
- iii. ...uncorrelated.
- iv. ...undefined.

- (b) What is the probability that the coin toss comes up heads, i.e, what is $P(H)$? **(4)**

1/2

15. Let X and Y be independent random variables, each with the Uniform(0, 1) distribution. Let $Z = \min(X, Y)$.

(a) Explain why

$$P(Z > z) = \begin{cases} 1, & \text{if } z \leq 0 \\ (1 - z)^2, & \text{if } 0 < z < 1 \\ 0, & \text{if } z \geq 1 \end{cases} \quad (4)$$

For $0 < z < 1$, $P(Z > z) = P(X > z, Y > z) = P(X > z)P(Y > z) = (1 - z)^2$.

(b) What is the probability density function (pdf) for Z ? (4)

$f_Z(z) = 2(1 - z)$ for $0 < z < 1$.

(c) What is the expected value of Z ? (4)

$1/3$

16. Suppose that (X, Y) has joint density

$$f_{XY}(x, y) = \begin{cases} \left(\frac{21}{4}\right) x^2 y, & \text{if } x^2 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Set up the integral to find the probability that X is larger than Y , but don't do the full calculation. Your final expression should **not** include the notation $f_{XY}(x, y)$. (4)

$$P(X > Y) = \int_0^1 \int_{x^2}^x \left(\frac{21}{4}\right) x^2 y \, dy \, dx$$

Useful Math Facts

1. **Results Regarding e :** For any number a ,

$$e^a = \sum_{i=0}^{\infty} \frac{a^i}{i!} \quad \text{and} \quad e^a = \lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n$$

2. **Geometric Series:** If $|r| < 1$ and a is an integer,

$$\sum_{k=a}^{\infty} r^k = \frac{r^a}{1-r}$$

3. **The Binomial Theorem:** For $k = 0, 1, 2, \dots$,

$$(\lambda_1 + \lambda_2)^k = \sum_{i=0}^k \binom{k}{i} \lambda_1^i \lambda_2^{k-i}$$

4. **The Gamma Function:**

$$\Gamma(t) = \int_0^{\infty} x^{t-1} e^{-x} dx$$

For any t , it holds that $\Gamma(t+1) = t\Gamma(t)$, and for $n = 1, 2, \dots$, $\Gamma(n) = (n-1)!$. Also, $\Gamma(0.5) = \sqrt{\pi}$.

5. **Definite Integrals:**

$$\int_0^{\infty} e^{-ax} dx = \frac{1}{a}, \quad \int_0^{\infty} x e^{-ax} dx = \frac{1}{a^2}, \quad \int_0^{\infty} x^m e^{-ax} dx = \frac{\Gamma(m+1)}{a^{m+1}}$$

$$\int_0^{\infty} e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}}, \quad \int_0^{\infty} x e^{-ax^2} dx = \frac{1}{2a}, \quad \int_0^{\infty} x^m e^{-ax^2} dx = \frac{\Gamma((m+1)/2)}{2a^{(m+1)/2}}$$

$$\int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \text{The Beta Function}$$

Table 1: Summary of some common distributions in probability and statistics.

Name	Density/Mass Function	MGF	Mean	Variance
Normal	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right),$ $-\infty < x < \infty, \sigma > 0$	$\exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$	μ	σ^2
Log-Normal	$\frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(x)-\mu)^2}{2\sigma^2}\right),$ $0 < x < \infty, \sigma > 0$	—	$\exp\left(\mu + \frac{\sigma^2}{2}\right)$	$e^{2(\mu+\sigma^2)} - e^{2\mu+\sigma^2}$
Gamma	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x},$ $0 < x < \infty, \alpha, \beta > 0$	$\left(\frac{\beta}{\beta-t}\right)^\alpha, \quad t < \beta$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$
Exponential	$\beta e^{-\beta x},$ $0 < x < \infty, \beta > 0$	$\frac{\beta}{\beta-t}, \quad t < \beta$	$\frac{1}{\beta}$	$\frac{1}{\beta^2}$
Beta	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1},$ $0 < x < 1, \alpha, \beta > 0$	$1 + \sum_{k=1}^{\infty} \left(\prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r}\right) \frac{t^k}{k!}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Uniform (continuous)	$\frac{1}{b-a},$ $a \leq x \leq b$	$\frac{e^{bt}-e^{at}}{t(b-a)}, \quad t \neq 0$ $1, \quad t = 0$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Bernoulli	$p^x(1-p)^{1-x},$ $x = 0, 1, 0 \leq p \leq 1$	$pe^t + (1-p)$	p	$p(1-p)$
Binomial	$\binom{n}{x} p^x (1-p)^{n-x},$ $x = 0, 1, \dots, n, 0 \leq p \leq 1$	$[pe^t + (1-p)]^n$	np	$np(1-p)$
Negative Binomial	$\binom{r+x-1}{x} p^r (1-p)^x,$ $x = 0, 1, \dots, r \in \{1, 2, \dots\}, 0 \leq p \leq 1$	$\left(\frac{p}{1-(1-p)e^t}\right)^r, \quad t < \log(1/(1-p))$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Geometric	$(1-p)^x p,$ $x = 0, 1, \dots, 0 \leq p \leq 1$	$\frac{p}{1-(1-p)e^t}, \quad t < \log(1/(1-p))$	$\frac{1-p}{p}$	$\frac{(1-p)}{p^2}$
Poisson	$\frac{e^{-\lambda} \lambda^x}{x!},$ $x = 0, 1, \dots, \lambda > 0$	$e^{\lambda(e^t-1)}$	λ	λ
Hypergeometric	$\frac{\binom{A}{x} \binom{B}{n-x}}{\binom{A+B}{n}},$ $x = 0, 1, \dots, n$	—	$\frac{nA}{A+B}$	$\frac{nAB(A+B-n)}{(A+B)^2(A+B+1)}$